

See the definitions of key terms in the glossary

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Solution

$$\begin{aligned}\text{heat change} &= m \times c \times \Delta T \\ &= 10.0 \text{ g} \times 0.385 \text{ J g}^{-1} \text{ }^{\circ}\text{C}^{-1} \times -60.0 \text{ }^{\circ}\text{C} \text{ (the value is negative as the Cu has lost heat)} \\ &= -231 \text{ J}\end{aligned}$$

Exercises

1 When a sample of NH_4SCN is mixed with solid $\text{Ba}(\text{OH})_2 \cdot 8\text{H}_2\text{O}$ in a glass beaker, the mixture changes to a liquid and the temperature drops sufficiently to freeze the beaker to the table. Which statement is true about the reaction?

- A The process is endothermic and ΔH is -
- B The process is endothermic and ΔH is +
- C The process is exothermic and ΔH is -
- D The process is exothermic and ΔH is +

2 Which one of the following statements is true of all exothermic reactions?

- A They produce gases.
- B They give out heat.
- C They occur quickly.
- D They involve combustion.

3 If 500 J of heat is added to 100.0 g samples of each of the substances below, which will have the largest temperature increase?

	Substance	Specific heat capacity / $\text{J g}^{-1} \text{K}^{-1}$
A	gold	0.129
B	silver	0.237
C	copper	0.385
D	water	4.18

4 The temperature of a 5.0 g sample of copper increases from $27 \text{ }^{\circ}\text{C}$ to $29 \text{ }^{\circ}\text{C}$. Calculate how much heat has been added to the system. (Specific heat capacity of $\text{Cu} = 0.385 \text{ J g}^{-1} \text{K}^{-1}$)

- A 0.770 J
- B 1.50 J
- C 3.00 J
- D 3.85 J

5 Consider the specific heat capacity of the following metals.

Metal	Specific heat capacity / $\text{J g}^{-1} \text{K}^{-1}$
Al	0.897
Be	1.82
Cd	0.231
Cr	0.449

1 kg samples of the metals at room temperature are heated by the same electrical heater for 10 min. Identify the metal which has the highest final temperature.

- A Al
- B Be
- C Cd
- D Cr

6 The specific heat of metallic mercury is $0.138 \text{ J g}^{-1} \text{ }^{\circ}\text{C}^{-1}$. If 100.0 J of heat is added to a 100.0 g sample of mercury at $25.0 \text{ }^{\circ}\text{C}$, what is the final temperature of the mercury?

Enthalpy changes and the direction of change

There is a natural direction for change. When we slip on a ladder, we go down, not up. The direction of change is in the direction of lower stored energy. In a similar way, we expect methane to burn when we strike a match and form carbon dioxide and water. The chemicals are changing in a way which reduces their enthalpy (Figure 5.5).

Animation

Select the icon to see a related animation



Quiz

Select the icon to take an interactive quiz to test your knowledge



CHALLENGE YOURSELF

2 Suggest an explanation for the pattern in specific heat capacities of the metals in Exercise 3.

Worked solutions

Select the icon at the end of the chapter to view worked solutions to exercises in this chapter

Answers

Select the icon at the end of the chapter to view answers to exercises in this chapter





01

Stoichiometric relationships

Essential ideas

- 1.1** Physical and chemical properties depend on the ways in which different atoms combine.
- 1.2** The mole makes it possible to correlate the number of particles with a mass that can be measured.
- 1.3** Mole ratios in chemical equations can be used to calculate reacting ratios by mass and gas volume.

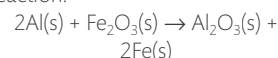
The birth of chemistry as a physical science can be traced back to the first successful attempts to quantify chemical change. Carefully devised experiments led to data that revealed one simple truth. *Chemical change involves interactions between particles that have fixed mass.* Even before knowledge was gained of the atomic nature of these particles and of the factors that determine their interactions, this discovery became the guiding principle for modern chemistry. We begin our study with a brief introduction to this particulate nature of matter, and go on to investigate some of the ways in which it can be quantified.

The term **stoichiometry** is derived from two Greek words – *stoicheion* for element and *metron* for measure. Stoichiometry describes the relationships between the amounts of reactants and products during chemical reactions. As it is known that matter is conserved during chemical change, stoichiometry is a form of book-keeping at the atomic level. It enables chemists to determine what amounts of substances they should react together and enables them to predict how much product will be obtained. The application of stoichiometry closes the gap between what is happening on the atomic scale and what can be measured.

In many ways this chapter can be considered as a toolkit for the mathematical content in much of the course. It covers the universal language of chemistry, **chemical equations**, and introduces the **mole** as the unit of amount. Applications include measurements of mass, volume, and concentration.

You may choose not to work through all of this at the start of the course, but to come back to these concepts after you have gained knowledge of some of the fundamental properties of chemical matter in Chapters 2, 3, and 4.

The reaction between ignited powdered aluminium and iron oxide, known as the thermite reaction:



Significant heat is released by the reaction, and it is used in welding processes including underwater welding. The stoichiometry of the reaction, as shown in the balanced equation, enables chemists to determine the reacting masses of reactants and products for optimum use.

1.1

Introduction to the particulate nature of matter and chemical change

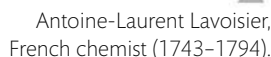
Understandings:

- Atoms of different elements combine in fixed ratios to form compounds, which have different properties from their component elements.

Guidance

Names and symbols of the elements are in the IB data booklet in Section 5.

- Mixtures contain more than one element and/or compound that are not chemically bonded together and so retain their individual properties.
- Mixtures are either homogeneous or heterogeneous.



Pictographic symbols used at the beginning of the 18th century to represent chemical elements and compounds. They are similar to those of the ancient alchemists. As more elements were discovered during the 18th century, attempts to devise a chemical nomenclature led to the modern alphabetic notational system. This system was devised by the Swedish chemist Berzelius and introduced in 1814.

Antoine-Laurent Lavoisier (1743–1794) is often called the ‘father of chemistry’. His many accomplishments include the naming of oxygen and hydrogen, the early development of the metric system, and a standardization of chemical nomenclature. Most importantly, he established an understanding of combustion as a process involving combination with oxygen from the air, and recognized that matter retains its mass through chemical change, leading to the law of conservation of mass. In addition, he compiled the first extensive list of elements in his book *Elements of Chemistry* (1789). In short, he changed chemistry from a qualitative to a quantitative science. But, as an unpopular tax collector in France during the French Revolution and Terror, he was tried for treason and guillotined in 1794. One and a half years after his death he was exonerated, and his early demise was recognized as a major loss to France.

The English language is based on an alphabet of just 26 letters. But, as we know, combining these in different ways leads to an almost infinite number of words, and then sentences, paragraphs, books, and so on. It is similar to the situation in chemistry, where the 'letters' are the single substances known as **chemical elements**. There are

4

only about 100 of these, but because of the ways in which they combine with each other, they make up the almost countless number of different chemical substances in our world.

In Chapter 2 we will learn about atomic structure, and how each element is made up of a particular type of atom. The atoms of an element are all the same as each other (with the exception of isotopes, which we will also discuss in Chapter 2), and are different from those of other elements. It is this distinct nature of its atoms that gives each element its individual properties. A useful definition of an atom is that it is the smallest particle of an element to show the characteristic properties of that element.

To help communication in chemistry, each element is denoted by a **chemical symbol** of either one upper case letter, or one upper case letter followed by a lower case letter. A few examples are given below.

Name of element	Symbol
carbon	C
fluorine	F
potassium	K
calcium	Ca
mercury	Hg
tungsten	W

You will notice that often the letter or letters used are derived from the English name of the element, but in some cases they derive from other languages. For example, Hg for mercury comes from Latin, whereas W for tungsten has its origin in European dialects. Happily, these symbols are all accepted and used internationally, so they do not need to be translated. A complete list of the names of the elements and their symbols is given in Section 5 of the IB data booklet.

Chemical compounds are formed from more than one element

Some elements, such as nitrogen and gold, are found in **native** form, that is uncombined with other elements in nature. But more commonly, elements exist in chemical combinations with other elements, in substances known as **chemical compounds**. Compounds contain a fixed proportion of elements, and are held together by chemical bonds (discussed in Chapter 4). The bonding between atoms in compounds changes their



Assorted minerals, including elements such as sulfur and silver, and compounds such as Al_2O_3 (sapphire) and CaF_2 (fluorite). Most minerals are impure and exist as mixtures of different elements and compounds.

The International Union of Pure and Applied Chemistry (IUPAC) was formed in 1919 by chemists from industry and academia. Since then it has developed into the world authority on chemical nomenclature and terminology. It has succeeded in fostering worldwide communications in the chemical sciences and in uniting academic, industrial, and public sector chemistry in a common language.

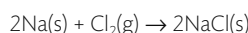
Chemistry is a very exact subject, and it is important to be careful in distinguishing between upper and lower case letters. For example, Co (cobalt, a metallic element) means something completely different from CO (carbon monoxide, a poisonous gas).

The number of elements that exist is open to change as new ones are discovered, although there is often a time-lag between a discovery and its confirmation by IUPAC. During this time a provisional systematic three-letter symbol is used, using Latin abbreviations to represent the atomic number. The letters u (un) = 1, b (bi) = 2, t (tri) = 3 and so on are used. So the provisional element of atomic number 118 will continue to be known as ununoctium or uuo until it is confirmed and a name formally agreed according to the process established by IUPAC.

A compound is a chemical combination of different elements, containing a fixed ratio of atoms. The physical and chemical properties of a compound are different from those of its component elements.



A combustion spoon holding sodium, Na, is lowered into a gas jar containing chlorine, Cl₂. The vigorous reaction produces white crystals of sodium chloride, NaCl.



The properties of the compound are completely different from those of its component elements.

A chemical equation shows:

direction of change

reactants $\longrightarrow$ products

Chemical equations are the universal language of chemistry. What other languages are universal, and to what extent do they help or hinder the pursuit of knowledge?

TOK

Figure 1.1 When hydrogen and oxygen react to form water, the atoms are rearranged, but the number of atoms of each element remains the same.

properties, so compounds have completely different properties from those of their component elements.

A classic example of this is that sodium, Na, is a dangerously reactive metal that reacts violently with water, while chlorine, Cl₂, is a toxic gas used as a chemical weapon. Yet when these two elements combine, they form the compound sodium chloride, NaCl, a white crystalline solid that we sprinkle all over our food.

Compounds are described using the chemical symbols for elements. A subscript is used to show the number of atoms of each element in a unit of the compound. Some examples are given below. (The reasons for the different ratios of elements in compounds will become clearer after we have studied atomic structure and bonding in Chapters 2 and 4.)

Name of compound	Symbol	Name of compound	Symbol
sodium chloride	NaCl	water	H ₂ O
potassium oxide	K ₂ O	glucose	C ₆ H ₁₂ O ₆
calcium bromide	CaBr ₂	ammonium sulfate	(NH ₄) ₂ SO ₄

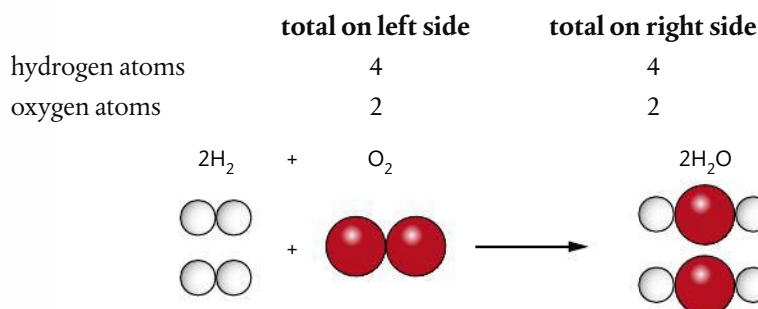
Chemical equations summarize chemical change

The formation of compounds from elements is an example of **chemical change** and can be represented by a **chemical equation**. A chemical equation is a representation using chemical symbols of the simplest ratio of atoms, as elements or in compounds, undergoing chemical change. The left-hand side shows the **reactants** and the right-hand side the **products**.

For example: calcium + chlorine $\rightarrow$ calcium chloride
 $\text{Ca} + \text{Cl}_2 \rightarrow \text{CaCl}_2$

As atoms are neither created nor destroyed during a chemical reaction, *the total number of atoms of each element must be the same on both sides of the equation*. This is known as **balancing the equation**, and uses numbers called **stoichiometric coefficients** to denote the number of units of each term in the equation.

For example: hydrogen + oxygen $\rightarrow$ water
 $2\text{H}_2 + \text{O}_2 \rightarrow 2\text{H}_2\text{O}$
 coefficients



Note that when the coefficient is 1, this does not need to be explicitly stated.

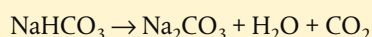
Chemical equations are used to show all types of reactions in chemistry, including reactions of decomposition, combustion, neutralization, and so on. Examples of these are given below and you will come across very many more during this course. Learning to write equations is an important skill in chemistry, which develops quickly with practice.

Worked example

Write an equation for the reaction of thermal decomposition of sodium hydrogencarbonate (NaHCO_3) into sodium carbonate (Na_2CO_3), water (H_2O), and carbon dioxide (CO_2).

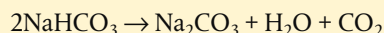
Solution

First write the information from the question in the form of an equation, and then check the number of atoms of each element on both sides of the equation.



	total on left side	total on right side
sodium atoms	1	2
hydrogen atoms	1	2
carbon atoms	1	2
oxygen atoms	3	6

In order to balance this we introduce coefficient 2 on the left.



Finally check that it is balanced for each element.



An equation by definition has to be *balanced*, so do not expect this to be specified in a question. After you have written an equation, *always* check the numbers of atoms of each element on both sides of the equation to ensure it is correctly balanced.



When a question refers to 'heating' a reactant or to 'thermal decomposition', this does not mean the addition of oxygen, only that heat is the source of energy for the reaction. If the question refers to 'burning' or 'combustion', this indicates that oxygen is a reactant.

NATURE OF SCIENCE

Early ideas to explain chemical change in combustion and rusting included the 'phlogiston' theory. This proposed the existence of a fire-like element that was released during these processes. The theory seemed to explain some of the observations of its time, although these were purely qualitative. It could not explain later quantitative data showing that substances actually gain rather than lose mass during burning. In 1783, Lavoisier's work on oxygen confirmed that combustion and rusting involve combination with oxygen from the air, so overturning the phlogiston theory. This is a good example of how the evolution of scientific ideas, such as how chemical change occurs, is based on the need for theories that can be tested by experiment. Where results are not compatible with the theory, a new theory must be put forward, which must then be subject to the same rigour of experimental test.



Exercises

- Write balanced chemical equations for the following reactions:
 - The decomposition of copper carbonate (CuCO_3) into copper(II) oxide (CuO) and carbon dioxide (CO_2).
 - The combustion of magnesium (Mg) in oxygen (O_2) to form magnesium oxide (MgO).
 - The neutralization of sulfuric acid (H_2SO_4) with sodium hydroxide (NaOH) to form sodium sulfate (Na_2SO_4) and water (H_2O).
 - The synthesis of ammonia (NH_3) from nitrogen (N_2) and hydrogen (H_2).
 - The combustion of methane (CH_4) to produce carbon dioxide (CO_2) and water (H_2O).
- Write balanced chemical equations for the following reactions:

(a) $\text{K} + \text{H}_2\text{O} \rightarrow \text{KOH} + \text{H}_2$	(b) $\text{C}_2\text{H}_5\text{OH} + \text{O}_2 \rightarrow \text{CO}_2 + \text{H}_2\text{O}$
(c) $\text{Cl}_2 + \text{KI} \rightarrow \text{KCl} + \text{I}_2$	(d) $\text{CrO}_3 \rightarrow \text{Cr}_2\text{O}_3 + \text{O}_2$
(e) $\text{Fe}_2\text{O}_3 + \text{C} \rightarrow \text{CO} + \text{Fe}$	



Remember when you are balancing an equation, change the stoichiometric coefficient but never change the subscript in a chemical formula.

When balancing equations, start with the most complex species, and leave terms that involve a single element to last.



Exercises

3 Use the same processes to balance the following examples:

- (a) $\text{C}_4\text{H}_{10} + \text{O}_2 \rightarrow \text{CO}_2 + \text{H}_2\text{O}$
- (b) $\text{NH}_3 + \text{O}_2 \rightarrow \text{NO} + \text{H}_2\text{O}$
- (c) $\text{Cu} + \text{HNO}_3 \rightarrow \text{Cu}(\text{NO}_3)_2 + \text{NO} + \text{H}_2\text{O}$
- (d) $\text{H}_2\text{O}_2 + \text{N}_2\text{H}_4 \rightarrow \text{N}_2 + \text{H}_2\text{O} + \text{O}_2$
- (e) $\text{C}_2\text{H}_7\text{N} + \text{O}_2 \rightarrow \text{CO}_2 + \text{H}_2\text{O} + \text{N}_2$



A chemical equation can be used to assess the efficiency of a reaction in making a particular product. The **atom economy** is a concept used for this purpose and is defined as:

$$\% \text{ atom economy} = \frac{\text{mass of desired product}}{\text{total mass of products}} \times 100$$

Note that this is different from % yield discussed later in this chapter, which is calculated using only one product and one reactant. Atom economy is an indication of how much of the reactants ends up in the required products, rather than in waste products. A higher atom economy indicates a more efficient and less wasteful process. The concept is increasingly used in developments in green and sustainable chemistry. This is discussed further in Chapters 12 and 13.

Mixtures form when substances combine without chemical interaction

Air is described as a **mixture** of gases because the separate components – different elements and compounds – are interspersed with each other, but are not chemically combined. This means, for example, that the gases nitrogen and oxygen when mixed

in air retain the same characteristic properties as when they are in the pure form. Substances burn in air because the oxygen present supports combustion, as does pure oxygen.

Another characteristic of mixtures is that their composition is not fixed. For example, air that we breathe in typically contains about 20% by volume oxygen, whereas the air that we breathe out usually contains only about 16% by volume oxygen. It is still correct to call both of these mixtures of air, because there is no fixed proportion in the definition.



A mixture is composed of two or more substances in which no chemical combination has occurred.



Tide of oily water heading towards Pensacola Beach, Florida, USA, after the explosion of the offshore drilling unit in the Gulf of Mexico in 2010. The oil forms a separate layer as it does not mix with the water.

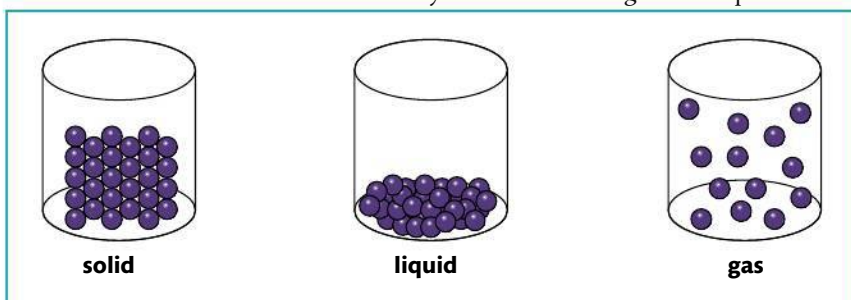
Air is an example of a **homogeneous mixture**, meaning that it has uniform composition and properties throughout. A solution of salt in water and a metal alloy such as bronze are also homogeneous. By contrast, a **heterogeneous mixture** such as water and oil has non-uniform composition, so its properties are not the same throughout. It is usually possible to see the separate components in a heterogeneous mixture but not in a homogeneous mixture.

Because the components retain their individual properties in a mixture, we can often separate them relatively easily. The technique we choose to achieve this will take advantage of a suitable difference in the physical properties of the components, as shown in the table below. Many of these are important processes in research and industry and are discussed in more detail in the following chapters.

Mixture	Difference in property of components	Technique used
sand and salt	solubility in water	solution and filtration
hydrocarbons in crude oil	boiling point	fractional distillation
iron and sulfur	magnetism	response to a magnet
pigments in food colouring	adsorption to solid phase	paper chromatography
different amino acids	net charge at a fixed pH	gel electrophoresis

Matter exists in different states determined by the temperature and the pressure

From our everyday experience, we know that all matter (elements, compounds, and mixtures) can exist in different forms depending on the temperature and the pressure. Liquid water changes into a solid form, such as ice, hail, or snow, as the temperature drops and it becomes a gas, steam, at high temperatures. These different forms are known as the **states of matter** and are characterized by the different energies of the particles.



- | | | |
|--|--|--|
| <ul style="list-style-type: none"> • particles close packed • inter-particle forces strong, particles vibrate in position • fixed shape • fixed volume | <ul style="list-style-type: none"> • particles more spaced • inter-particle forces weaker, particles can slide over each other • no fixed shape • fixed volume | <ul style="list-style-type: none"> • particles spread out • inter-particle forces negligible, particles move freely • no fixed shape • no fixed volume |
|--|--|--|



Ocean oil spills are usually the result of accidents in the industries of oil extraction or transport. The release of significant volumes of oil causes widespread damage to the environment, especially wildlife, and can have a major impact on local industries such as fishing and tourism. Efforts to reduce the impact of the spill include the use of dispersants, which act somewhat like soap in helping to break up the oil into smaller droplets so it can mix better with water. Concern is expressed, however, that these chemicals may increase the toxicity of the oil and they might persist in the environment. The effects of an oil spill often reach countries far from the source and are the subject of complex issues in international law. With the growth in demand for offshore drilling for oil and projected increases in oil pipelines, these issues are likely to become all the more pressing.

Figure 1.2 Representation of the arrangement of the particles of the same substance in the solid, liquid, and gas states.



Depending on the chemical nature of the substance, matter may exist as atoms such as Ar(g) , or as molecules such as $\text{H}_2\text{O(l)}$, or as ions such as Na^+ and Cl^- in NaCl(aq) . The term **particle** is therefore used as an inclusive term that is applied in this text to any or all of these entities of matter.

Temperature is a measure of the average kinetic energy of the particles of a substance.



Bromine liquid, $\text{Br}_2(\text{l})$, had been placed in the lower gas jar only, and its vapour has diffused to fill both jars. Bromine vaporizes readily at room temperature and the gas colour allows the diffusion to be observed. Because gas molecules can move independently of each other and do so randomly, a gas spreads out from its source in this way.

This is known as the **kinetic theory** of matter. It recognizes that the average kinetic energy of the particles is directly related to the temperature of the system. The state of matter at a given temperature and pressure is determined by the strength of forces that may exist between the particles, known as **inter-particle forces**. The average kinetic energy is proportional to the temperature in Kelvin, introduced on page 37.

Worked example

Which of the following has the highest average kinetic energy?

A He at 100°C

B H_2 at 200°C

C O_2 at 300°C

D H_2O at 400°C

Solution

Answer = D. The substance at the highest temperature has the highest average kinetic energy.

Liquids and gases are referred to as **fluids**, which refers to their ability to flow. In the case of liquids it means that they take the shape of their container. Fluid properties are why **diffusion** occurs predominantly in these two states. Diffusion is the process by which the particles of a substance become evenly distributed, as a result of their random movements.

Kinetic energy (KE) refers to the energy associated with movement or motion. It is determined by the mass (m) and velocity or speed (v) of a substance, according to the relationship:

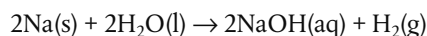
$$\text{KE} = \frac{1}{2}mv^2$$

As the kinetic energy of the particles of substances at the same temperature is equal, this means there is an inverse relationship between mass and velocity. This is why substances with lower mass diffuse more quickly than those with greater mass, when measured at the same temperature. This is discussed in more detail in Chapter 14.

State symbols are used to show the states of the reactants and products taking part in a reaction. These are abbreviations, which are given in brackets after each term in an equation, as shown below.

State	Symbol	Example
solid	(s)	$\text{Mg}(\text{s})$
liquid	(l)	$\text{Br}_2(\text{l})$
gas	(g)	$\text{N}_2(\text{g})$
aqueous (dissolved in water)	(aq)	$\text{HCl}(\text{aq})$

For example:



Exercises

4 Classify the following mixtures as homogeneous or heterogeneous:

(a) sand and water

(b) smoke

(c) sugar and water

(d) salt and iron filings

(e) ethanol and water in wine

(f) steel

Exercises

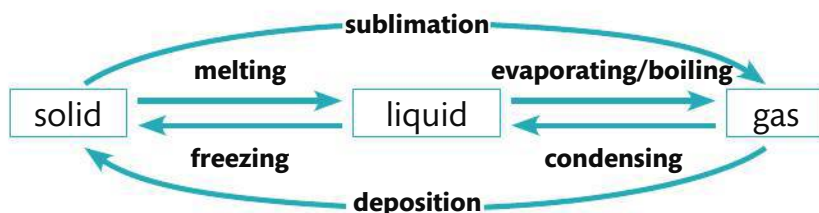
- 5 Write balanced equations for the following reactions and apply state symbols to all reactants and products, assuming room temperature and pressure unless stated otherwise. If you are not familiar with the aqueous solubilities of some of these substances, you may have to look them up.
- (a) $\text{KNO}_3 \rightarrow \text{KNO}_2 + \text{O}_2$ (when heated, 500°C)
 - (b) $\text{CaCO}_3 + \text{H}_2\text{SO}_4 \rightarrow \text{CaSO}_4 + \text{CO}_2 + \text{H}_2\text{O}$
 - (c) $\text{Li} + \text{H}_2\text{O} \rightarrow \text{LiOH} + \text{H}_2$
 - (d) $\text{Pb}(\text{NO}_3)_2 + \text{NaCl} \rightarrow \text{PbCl}_2 + \text{NaNO}_3$ (all reactants are in aqueous solution)
 - (e) $\text{C}_3\text{H}_6 + \text{O}_2 \rightarrow \text{CO}_2 + \text{H}_2\text{O}$ (combustion reaction)
- 6 A mixture of two gases, X and Y, which both have strong but distinct smells, is released. From across the room the smell of X is detected more quickly than the smell of Y. What can you deduce about X and Y?
- 7 Ice floats on water. Comment on why this is not what you would expect from the kinetic theory of matter.



It is good practice to show state symbols in all equations, even when they are not specifically requested.

Matter changes state reversibly

As the movement or kinetic energy of the particles increases with temperature, they will overcome the inter-particle forces and change state. These state changes occur at a fixed temperature and pressure for each substance, and are given specific names shown below.



Sublimation, the direct inter-conversion of solid to gas without going through the liquid state, is characteristic at atmospheric pressure of some substances such as iodine, carbon dioxide, and ammonium chloride. Deposition, the reverse of sublimation that changes a gas directly to solid, is responsible for the formation of snow, frost, and hoar frost.

Note that evaporation involves the change of liquid to gas, but, unlike boiling, evaporation occurs only at the surface and takes place at temperatures below the boiling point.



Drying clothes. The heat of the Sun enables all the water to evaporate from the clothes.

Ice crystals, known as Hair Ice, formed by deposition on dead wood in a forest on Vancouver Island, Canada.

Essential ideas

- | | |
|------|---|
| 8.1 | Many reactions involve the transfer of a proton from an acid to a base. |
| 8.2 | The characterization of an acid depends on empirical evidence such as the production of gases in reactions with metals, the colour changes of indicators, or the release of heat in reactions with metal oxides and hydroxides. |
| 8.3 | The pH scale is an artificial scale used to distinguish between acid, neutral, and basic/alkaline solutions. |
| 8.4 | The pH depends on the concentration of the solution. The strength of acids or bases depends on the extent to which they dissociate in aqueous solution. |
| 18.1 | The acid–base concept can be extended to reactions that do not involve proton transfer. |
| 18.2 | The equilibrium law can be applied to acid–base reactions. Numerical problems can be simplified by making assumptions about the relative concentrations of the species involved. The use of logarithms is also significant here. |
| 18.3 | pH curves can be investigated experimentally but are mathematically determined by the dissociation constants of the acid and base. An indicator with an appropriate end point can be used to determine the equivalence point of the reaction. |
| 8.5 | Increased industrialization has led to greater production of nitrogen and sulfur oxides leading to acid rain, which is damaging our environment. These problems can be reduced through collaboration with national and intergovernmental organizations. |

Acid–base theory informs much of our research into rainwater pollution and other aspects of environmental change.

The burning feeling of acid indigestion, the sour taste of grapefruit, and the vinegary smell of wine that has been exposed to the air are just some of the everyday encounters we have with acids. Likewise alkalis, or bases, are familiar substances – for example in baking soda, in household cleaners that contain ammonia, and in medication against indigestion. So what are the defining properties of these two groups of substances?

This question has intrigued chemists for centuries. The word ‘acid’ is derived from the Latin word *acetum* meaning sour – early tests to determine whether a substance was acidic were based on tasting! But it was learned that acids had other properties in common too: for example they changed the colour of the dye litmus from blue to red and they corroded metals. Similarly, alkalis were known to have distinctive properties such as being slippery to the touch, being able to remove fats and oils from fabrics, and turning litmus from red to blue. The name alkali comes from the Arabic word for plant ash, *alkalja*, where they were first identified. Early theories about acids and alkalis focused only on how they reacted together – it was actually suggested that the sourness of acids came from their possession of sharp angular spikes which became embedded in soft, rounded particles of alkali!



Over the last 120 years, our interpretation of acid–base behaviour has evolved alongside an increasing knowledge of atomic structure and bonding. In this chapter, we explore the modern definitions of acids and bases and learn how these help us to interpret and predict their interactions. Acid–base theory is central to topics such as air and water pollution, how global warming may affect the chemistry of the oceans, the action of drugs in the body, and many other aspects of cutting-edge research. As most acid–base reactions involve equilibria, much of the approach and mathematical content is based on work covered in Chapter 7. It is strongly recommended that you are familiar with this work before starting this chapter.

◀ A food scientist testing a sample of a batch of orange juice in a factory in France. pH measurements are an important part of the quality control of the product.

8.1 Theories of acids and bases

Understandings:

- A Brønsted–Lowry acid is a proton/ H^+ donor and a Brønsted–Lowry base is a proton/ H^+ acceptor.

Guidance

Students should know the representation of a proton in aqueous solution as both $H^+(aq)$ and $H_3O^+(aq)$.

- Amphiprotic species can act as both Brønsted–Lowry acids and bases.
- A pair of species differing by a single proton is called a conjugate acid–base pair.

Applications and skills:

- Deduction of the Brønsted–Lowry acid and base in a chemical reaction.

Guidance

The location of the proton transferred should be clearly indicated. For example, CH_3COOH/CH_3COO^- rather than $C_2H_4O_2/C_2H_3O_2^-$.

- Deduction of the conjugate acid or conjugate base in a chemical reaction.

Guidance

Lewis theory is not required here.

Early theories

The famous French chemist Lavoisier proposed in 1777 that oxygen was the ‘universal acidifying principle’. He believed that an acid could be defined as a compound of oxygen and a non-metal. In fact, the name he gave to the newly discovered gas *oxygen* means ‘acid-former’. This theory, however, had to be dismissed when the acid HCl was proven to be made of hydrogen and chlorine only – no oxygen. To hold true, of course, any definition of an acid has to be valid for *all* acids.

A big step forward came in 1887 when the Swedish chemist Arrhenius suggested that an acid could be defined as a substance that dissociates in water to form hydrogen ions (H^+) and anions, while a base dissociates into hydroxide (OH^-) ions and cations. He also recognized that the hydrogen and hydroxide ions could form water, and the cations and anions form a salt. In a sense, Arrhenius was very close to the theory that is widely used to explain acid and base properties today, but his focus was only on aqueous systems. A broader theory was needed to account for reactions occurring without water, and especially for the fact that some insoluble substances show base properties.

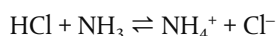
Svante August Arrhenius (1859–1927) wrote up his ideas on acids dissociating into ions in water as part of his doctoral thesis while a student at Stockholm University. But his theory was not well received and he was awarded the lowest possible class of degree. Later, his work gradually gained recognition and he received one of the earliest Nobel Prizes in Chemistry in 1903.

Arrhenius may be less well known as the first person documented to predict the possibility of global warming as a result of human activity. In 1896, aware of the rising levels of CO_2 caused by increased industrialization, he calculated the likely effect of this on the temperature of the Earth. Today, over 100 years later, the significance of this relationship between increasing CO_2 and global temperatures has become a subject of major international concern.

Brønsted–Lowry: a theory of proton transfer

In 1923 two chemists, Martin Lowry of Cambridge, England, and Johannes Brønsted of Copenhagen, Denmark, working independently, published similar conclusions regarding the definitions of acids and bases. Their findings overcame the limitations of Arrhenius' work and have become established as the **Brønsted–Lowry theory**.

This theory focuses on the transfer of H^+ ions during an acid–base reaction: acids donate H^+ while bases accept H^+ . For example, in the reaction between HCl and NH_3 :



HCl transfers H^+ to NH_3 and so acts as an acid; NH_3 accepts the H^+ and so acts as a base.

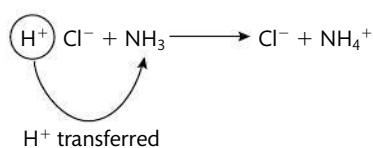


Figure 8.1 HCl transfers H^+ to NH_3 .

Hydrogen atoms contain just one proton and one electron, so when they ionize by losing the electron, all that is left is the proton. Therefore H^+ is *equivalent to a proton*, and we will use the two terms interchangeably here.

The Brønsted–Lowry theory can therefore be stated as:

- a Brønsted–Lowry acid is a proton (H^+) donor;
- a Brønsted–Lowry base is a proton (H^+) acceptor.

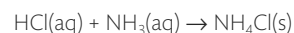


NATURE OF SCIENCE

The evolution of theories to explain acid–base chemistry and develop general principles is a fascinating tale of the scientific process in action. Some theories have arisen and been disproved, such as Lavoisier's early definition of an acid as a compound containing oxygen. Falsification of an idea when it cannot be applied in all cases is an essential aspect of the scientific process. Other theories have proved to be too limited in application, such as Arrhenius' theory which could not be generalized beyond aqueous solutions.

On the other hand, Brønsted–Lowry theory (as well as an alternate broader theory known as Lewis theory that is described later), has stood the test of time and experimentation. This indicates that it has led to testable predictions which have supported the theory, and enabled wide ranging applications to be made.

Reaction between vapours of HCl and NH_3 forming the white smoke of ammonium chloride NH_4Cl .





NATURE OF SCIENCE

Brønsted and Lowry's work on acid-base theory may be a good example of what is sometimes referred to as 'multiple independent discovery'. This refers to cases where similar discoveries are made by scientists at almost the same time, even though they have worked independently from each other. Examples include the independent discovery of the calculus by Newton and Leibniz, the formulation of the mechanism for biological evolution by Darwin and Wallace, and derivation of the pressure-volume relationships of gases by Boyle and Mariotte. Nobel Prizes in Chemistry are commonly awarded to more than one person working in the same field, who may have made the same discovery independently. Multiple independent discoveries are likely increasing as a result of communication technology, which enables scientists who may be widely separated geographically to have access to a common body of knowledge as the basis for their research.

When a discovery is made by more than one scientist at about the same time, ethical questions can arise, such as whether the credit belongs to one individual or should be shared, and whether there are issues regarding intellectual property and patent rights. Historically, these issues have been settled in very different ways in different cases. Nonetheless, the peer review process in science aims to ensure that credit is correctly awarded to the scientist or scientists responsible for a discovery, and that published work represents a new contribution to work in that field.

A Brønsted–Lowry acid is a proton (H^+) donor.

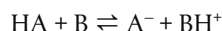
A Brønsted–Lowry base is a proton (H^+) acceptor.



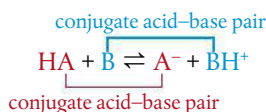
Conjugate pairs

The act of donating cannot happen in isolation – there must always be something present to play the role of acceptor. In Brønsted–Lowry theory, an acid can therefore only behave as a proton donor if there is also a base present to accept the proton.

Let's consider the acid–base reaction between a generic acid HA and base B:

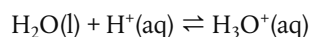


We can see that HA acts as an acid, donating a proton to B while B acts as a base, accepting the proton from HA. But if we look also at the reverse reaction, we can pick out another acid–base reaction: here BH^+ is acting as an acid, donating its proton to A^- while A^- acts as a base accepting the proton from BH^+ . In other words acid HA has reacted to form the base A^- , while base B has reacted to form acid BH^+ .



Acids react to form bases and vice versa. The acid–base pairs related to each other in this way are called **conjugate acid–base pairs**, and you can see that they *differ by just one proton*. It is important to be able to recognize these pairs in a Brønsted–Lowry acid–base reaction.

One example of a conjugate pair is H_2O and H_3O^+ , which is found in all acid–base reactions in aqueous solution. The reaction



occurs when a proton released from an acid readily associates with H_2O molecules forming H_3O^+ . In other words, protons become hydrated. H_3O^+ is variously called the hydroxonium ion, the oxonium ion, or the hydronium ion and is always the form of hydrogen ions in aqueous solution. However, for most reactions it is convenient simply to write it as $\text{H}^+(\text{aq})$. Note that in this pair H_3O^+ is the conjugate acid and H_2O its conjugate base.

Note that $\text{H}_3\text{O}^+(\text{aq})$ and $\text{H}^+(\text{aq})$ are both used to represent a proton in aqueous solution.



Lowry described the ready hydration of the proton as 'the extreme reluctance of the hydrogen nucleus to lead an isolated existence'.



Essential ideas

9.1

Redox (reduction–oxidation) reactions play a key role in many chemical and biochemical processes.

9.2
and
19.1

Voltaic cells convert chemical energy to electrical energy and electrolytic cells convert electrical energy to chemical energy. Energy conversions between electrical and chemical energy lie at the core of electrochemical cells.

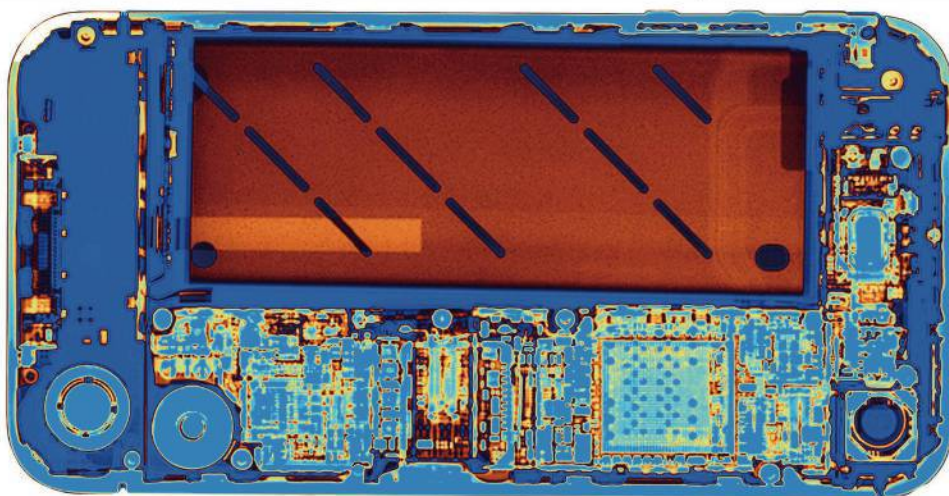
Oxygen makes up only about 20% by volume of the air, yet it is the essential component for so many reactions. Without it fuels would not burn, iron would not rust, and we would be unable to obtain energy from our food molecules through respiration. Indeed, animal life on the planet did not evolve until a certain concentration of oxygen had built up in the atmosphere over 600 million years ago. The term **oxidation** has been in use for a long time to describe these and other reactions where oxygen is involved in chemical change. Oxidation, though, is only half of the story, as it is always accompanied by the opposite process **reduction**, which was originally thought of in terms of loss of oxygen.

Later, however, the terms widened to include a much broader range of reactions. We now define these two processes, oxidation and reduction, as occurring whenever electrons are transferred from one reactant to another – and many of these reactions do not use oxygen. For example, photosynthesis, the process by which plants store chemical energy from light energy, involves oxidation and reduction reactions, although oxygen itself is not a reactant.

Transferring electrons from one substance to another leads to a flow of electrons, in other words an electric current. So, chemical reactions can be used to generate electricity – a simple **voltaic cell** or **battery** works in this way. By reversing the process and using an electric current to drive reactions of oxidation and reduction, stable compounds can be broken down into their elements. This is the process known as **electrolysis**. These applications of oxidation and reduction, collectively known as **electrochemical cells**, have truly revolutionized our world. For example, it would be hard to imagine life without the many battery-powered mobile devices we use every day, or without metals such as aluminium that are only available from electrolysis. An understanding of oxidation and reduction is therefore at the heart of the study of a large branch of chemistry both in the laboratory and beyond.

The vigorous reaction here shows the redox change that occurs when magnesium metal and copper oxide are heated together, forming magnesium oxide and copper. $\text{Mg(s)} + \text{CuO(s)} \rightarrow \text{MgO(s)} + \text{Cu(s)}$. Redox reactions occur when electrons are transferred between species, and are the basis of electrochemistry.

Coloured X-ray of a smart phone, showing its rechargeable battery. Mobile devices such as this are powered by electrical energy generated from the reactions of oxidation and reduction occurring within the battery.



9.1 Oxidation and reduction

Understandings:

- Oxidation and reduction can be considered in terms of oxygen gain/hydrogen loss, electron transfer, or change in oxidation number.
- An oxidizing agent is reduced and a reducing agent is oxidized.

Guidance

Oxidation states should be represented with the sign given before the number, e.g. +2 not 2+.

- Variable oxidation numbers exist for transition metals and for most main-group non-metals.

Guidance

The oxidation state of hydrogen in metal hydrides (-1) and oxygen in peroxides (-1) should be covered.

- The activity series ranks metals according to the ease with which they undergo oxidation.

Guidance

A simple activity series is given in the IB data booklet in section 25.

- The Winkler method can be used to measure biochemical oxygen demand (BOD), used as a measure of the degree of pollution in a water sample.

Applications and skills:

- Deduction of the oxidation state of an atom in an ion or a compound.
- Deduction of the name of a transition metal compound from a given formula, applying oxidation numbers represented by Roman numerals.

Guidance

Oxidation number and oxidation state are often used interchangeably, though IUPAC does formally distinguish between the two terms. Oxidation numbers are represented by Roman numerals according to IUPAC.

- Identification of the species oxidized and reduced and the oxidizing and reducing agents, in redox reactions.
- Deduction of redox reactions using half-equations in acidic or neutral solutions.
- Deduction of the feasibility of a redox reaction from the activity series or reaction data.
- Solution of a range of redox titration problems.
- Application of the Winkler method to calculate BOD.

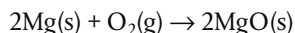
Introduction to oxidation and reduction

Early attempts to define oxidation and reduction were based on observations of the gain and loss of oxygen and hydrogen during chemical change. They can be summarized as follows.

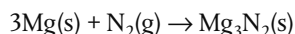
Oxidation	Reduction
gain of oxygen	loss of oxygen
loss of hydrogen	gain of hydrogen

These definitions are still useful in some cases, particularly in organic chemistry, but they have largely been replaced by a more inclusive approach. It is now recognized that oxidation and reduction occur during chemical change whenever there is a shift in electron density from one atom to another. In other words, the focus has become the transfer of electrons, whether complete or partial, whether oxygen and hydrogen are involved or not. Some examples will help to explain this.

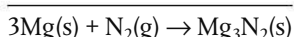
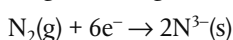
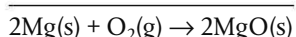
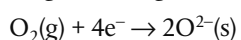
When magnesium is burned in air, it gives a bright white flame with the production of a white powder, magnesium oxide:



The fact that magnesium gains oxygen in this reaction makes it easy to see why we say it is an **oxidation** reaction and that magnesium has been oxidized. However, during the same reaction a small amount of the magnesium combines with the nitrogen of the air too, forming magnesium nitride. It may be less obvious that this is also an oxidation reaction and that again magnesium is oxidized.



What do these two reactions have in common which means they can both be defined in this way? If we divide them into so-called **half-equations**, each showing what happens to one reactant, we can examine what is happening in terms of electrons.



In both reactions Mg is forming Mg^{2+} by losing electrons, while O and N are forming O^{2-} and N^{3-} respectively by gaining electrons. It is this transfer of electrons that now defines oxidation, and its opposite reaction, reduction.

🔥 Oxidation is the loss of electrons, reduction is the gain of electrons. 🍷

So in the reactions above, magnesium is oxidized while oxygen and nitrogen are reduced. Clearly each process is dependent on the other, so oxidation and reduction will always occur together and reactions of this type are known as **redox reactions**.

The free-radical theory of ageing suggests that the physiological changes associated with ageing are the result of oxidative reactions in cells causing damage to membranes and large molecules such as DNA. These changes accumulate with time and may explain the increase in degenerative diseases, such as cancer, with age. The theory suggests that supplying cells with anti-oxidants will help to slow down the damaging oxidative reactions. Anti-oxidants are particularly abundant in fresh fruit and vegetables, as well as in red wine, tea, and cocoa. Although there is strong evidence that anti-oxidant supplementation may help protect against certain diseases, it has not yet been shown to produce a demonstrated increase in the human life span.



▲ Fresh fruits and vegetables are good sources of anti-oxidants, which may help prevent damaging oxidative reactions in cells.

Anti-oxidants are particularly abundant in fresh fruit and vegetables, as well as in red wine, tea, and cocoa. Although there is strong evidence that anti-oxidant supplementation may help protect against certain diseases, it has not yet been shown to produce a demonstrated increase in the human life span.



▲ Magnesium ribbon burning in air. It forms mostly magnesium oxide with small amounts of magnesium nitride.



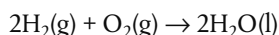
It may help you to remember OILRIG: Oxidation Is Loss (of electrons), Reduction Is Gain (of electrons).



Oxidation is the loss of electrons, reduction is the gain of electrons.

Oxidation numbers enable us to track redox change

In reactions involving ions, such as the examples above, it is easy to identify the electron transfers occurring. But what about reactions where electrons are not transferred but instead are shared in a covalent bond, such as the combination of hydrogen and oxygen to produce water? Can oxidation and reduction be identified here too?



The answer is 'yes', through the introduction of the concept of **oxidation state**. This is a value we assign to each atom in a compound that is a measure of the electron control or possession it has relative to the atom in the pure element. Oxidation state can be defined as the apparent charge that an atom has in a molecule or ion. It is as if we exaggerate the unequal sharing in a covalent bond, based on electronegativity differences, to the point where each atom has complete gain or loss of the electrons shared – in other words, has formed ions. This enables us to keep track of the relative electron density in a compound and how it changes during a reaction. There are two parts to the oxidation state:

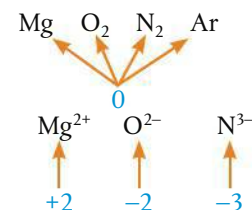
- the sign: '+' means the atom has lost electron control, '-' means it has gained electron control;
- its value: this refers to the number of electrons over which control has changed.

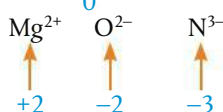
The oxidation state is written with the sign first, followed by the number, e.g. +2 or -3.

Strategy for assigning oxidation states

There are a few simple rules to follow which help in determining the oxidation state of an atom in any species. We will use blue to show oxidation states throughout this text.

- 1 Atoms in the free (uncombined) element have an oxidation state of zero: for example,


- 2 In simple ions, the oxidation state is the same as the charge on the ion: for example,


- 3 The oxidation states of all the atoms in a neutral (uncharged) compound must add up to zero:

for example, in H_2SO_4 the sum of oxidation states = 0.
- 4 The oxidation states of all the atoms in a polyatomic ion must add up to the charge on the ion:

for example, in SO_4^{2-} the sum of oxidation states = -2.
- 5 The usual oxidation state for an element is the same as the charge on its most common ion:

for example, Group 1 elements have oxidation state = +1, H is usually +1, O is usually -2.
- 6 Most main group non-metals, the elements at the bottom of Group 14, and transition elements have oxidation states that vary in different compounds – depending on the conditions and other elements present. So for N, P, S, Sn, Pb, and all transition elements, the oxidation state of the element in a particular species needs to be determined on a case-by-case basis.

The table on page 409 summarizes some useful information about oxidation states.

Note that the charge on an ion X is written with the number first then the charge: e.g. X^{2+} . The oxidation state is written with the charge first then the number: e.g. +2.



What ways of knowing can we use to distinguish between the charge on an ion and the oxidation state?

TOK

Element	Usual oxidation state	Exceptions	Explanation
Li, Na, K	+1		
Mg, Ca	+2		
F	-1		there are no exceptions because F is the most electronegative element
O	-2	peroxides such as H_2O_2 , where it is -1; OF_2 , where it is +2	
H	+1	metal hydrides such as NaH , where it is -1	H is more electronegative than Na and so gains electron control
Cl	-1	when it is combined with O or F	Cl is less electronegative than O and F, and so loses electron control

When working out the oxidation states of atoms in compounds, it is usually best to assign the oxidation state to the atoms that are easy to predict first, then use Rules 3 and 4 above to find the more unpredictable elements by subtraction.

Worked example

Assign oxidation states to all the elements in (a) H_2SO_4 and (b) SO_3^{2-} .

Solution

(a) We can assign H and O as follows:

$$\begin{array}{c} \text{H}_2\text{SO}_4 \\ \uparrow \quad \uparrow \\ +1 \quad -2 \end{array}$$

Note that the oxidation states apply to each atom and that here the sum of all the oxidation states must be zero as H_2SO_4 is electrically neutral.

Therefore, $2(+1) + \text{S} + 4(-2) = 0 \Rightarrow \text{S} = +6$

(b) Here we start by assigning O:

$$\begin{array}{c} \text{SO}_3^{2-} \\ \uparrow \\ -2 \end{array}$$

Note that here the oxidation states must add up to -2, the charge on the ion.

Therefore, $\text{S} + 3(-2) = -2 \Rightarrow \text{S} = +4$

The oxidation state of a transition metal in a complex ion can be worked out from the charges on the ligands and the overall charge. Remember from Chapter 3 that ligands are either neutral or negatively charged.

Worked example

Assign oxidation states to the metal ion in (a) $[\text{Co}(\text{NH}_3)_6]^{3+}$ and (b) $[\text{CuCl}_4]^{2-}$.

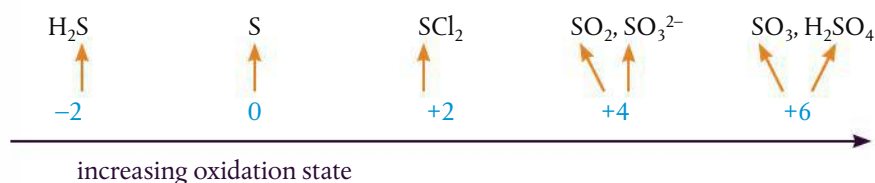
Solution

(a) NH_3 is a neutral ligand, so the charge on the complex is the same as the charge on the metal ion: $\therefore \text{Co} = +3$

(b) Cl has a 1- charge $\therefore (\text{charge on Cu}) + (4 \times 1-) = 2-$
 $\therefore \text{Cu} = +2$

Interpreting oxidation states

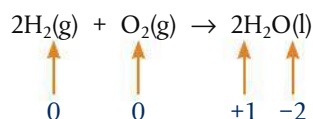
We can see that an element such as sulfur can have a wide range of oxidation states in different compounds.



What is the significance of these different values? Because the oxidation state is a measure of the electron control that an atom has, it follows that the higher the positive number, the more the atom has lost control over electrons, in other words the more oxidized it is. Likewise, the greater the negative number, the more it has gained electron control, so the more reduced it is. Therefore H_2S represents sulfur in its most reduced form (lowest oxidation number), and SO_3 and H_2SO_4 represent sulfur in its most oxidized form (highest oxidation number).

It follows that any change in oxidation states during a reaction is an indication that redox processes are occurring: an increase in oxidation number represents oxidation, and a decrease in oxidation number represents reduction.

So, going back to the reaction between hydrogen and oxygen discussed earlier, we can now clearly follow the redox process:



Hydrogen has been oxidized (oxidation state increased from 0 to +1) and oxygen has been reduced (oxidation state decreased from 0 to -2).

A redox reaction is a chemical reaction in which changes in the oxidation states occur.

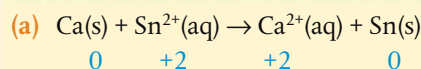
Oxidation occurs when there is an increase in oxidation state of an element, reduction occurs when there is a decrease in oxidation state of an element.

Worked example

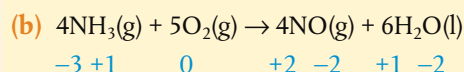
Use oxidation states to deduce which species is oxidized and which is reduced in the following reactions:

- (a) $\text{Ca}(\text{s}) + \text{Sn}^{2+}(\text{aq}) \rightarrow \text{Ca}^{2+}(\text{aq}) + \text{Sn}(\text{s})$
 (b) $4\text{NH}_3(\text{g}) + 5\text{O}_2(\text{g}) \rightarrow 4\text{NO}(\text{g}) + 6\text{H}_2\text{O}(\text{l})$

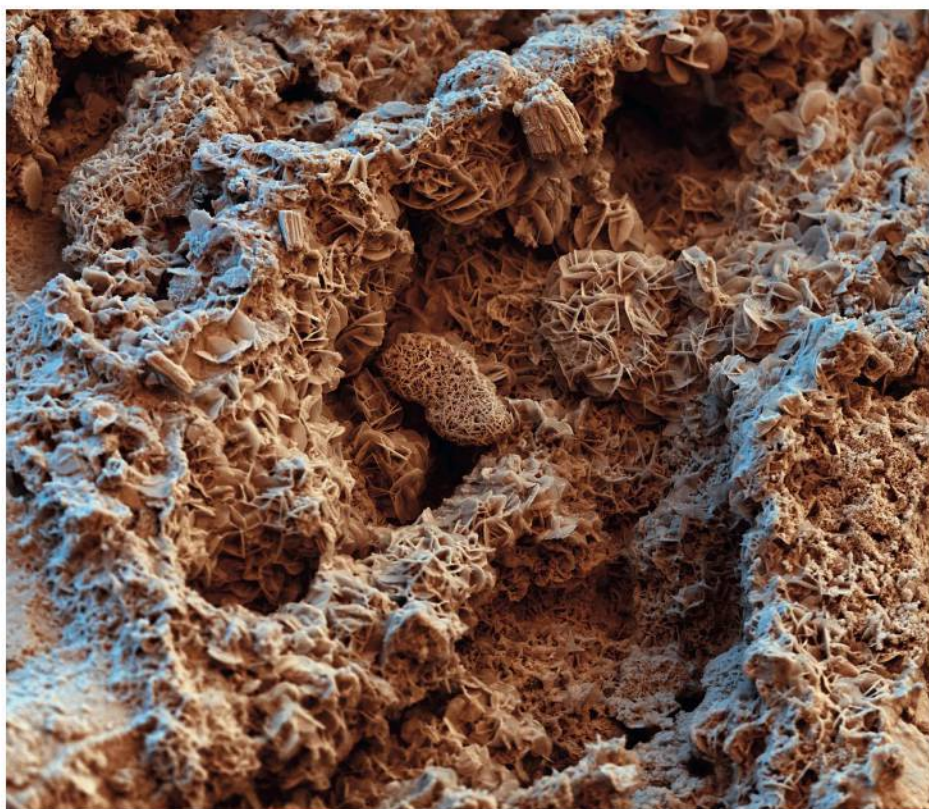
Solution



Ca is oxidized because its oxidation state increases from 0 to +2; Sn^{2+} is reduced because its oxidation state decreases from +2 to 0.



NH_3 is oxidized because the oxidation state of N increases from -3 to +2; O_2 is reduced because its oxidation state decreases from 0 to -2.



Coloured scanning electron micrograph (SEM) of rust on painted sheet metal. Rust is hydrated iron oxide resulting from electrochemical reactions between iron and atmospheric water vapour and oxygen. Rusting spontaneously causes iron to revert to its more stable oxidized state, unless this is prevented by protecting the iron. Rust flakes off the surface of the iron, causing it to be degraded and weakened.

Systematic names of compounds use oxidation numbers

We have seen that elements such as sulfur exhibit different oxidation states in different compounds. In these cases it is useful to include information about the oxidation state in the name. Traditional names used descriptive language that became associated with a particular oxidation state. For example, ferrous and ferric iron oxides referred to FeO and Fe_2O_3 , in which Fe has oxidation states **+2** and **+3** respectively.

The IUPAC system, founded in 1919, introduced a nomenclature using **oxidation numbers** to make the names more recognizable and unambiguous. This system uses a Roman numeral corresponding to the oxidation state which is inserted in brackets after the name of the element. For example copper(I) oxide contains copper in the oxidation state **+1**. Different compounds in which the same elements have different oxidation states are then shown with different oxidation numbers, such as copper(I) oxide and copper(II) oxide.

The table shows some common examples.

Formula of compound	Oxidation state	Name using oxidation number	Formula of compound	Oxidation state	Name using oxidation number
FeO	Fe +2	iron(II) oxide	MnO_4^-	Mn +7	manganate(VII) ion
Fe_2O_3	Fe +3	iron(III) oxide	$\text{K}_2\text{Cr}_2\text{O}_7$	Cr +6	potassium dichromate(VI)
Cu_2O	Cu +1	copper(I) oxide	Cr_2O_3	Cr +3	chromium(III) oxide
CuO	Cu +2	copper(II) oxide	NO	N +2	nitrogen(II) oxide
MnO_2	Mn +4	manganese(IV) oxide	NO_2	N +4	nitrogen(IV) oxide



Oxidation state is shown with a '+' or '-' sign and an Arabic numeral, e.g. **+2**.

Oxidation number is shown by inserting a Roman numeral in brackets after the name or symbol of the element.

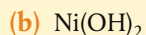
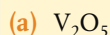
Although we have made the distinction here between oxidation number and oxidation state according to IUPAC guidelines, in reality this is not the main point of this discussion. In fact many sources use oxidation number and oxidation state interchangeably, so it is not too important. What does matter though, is the information represented in these values in terms of the electron density associated with an atom in a compound.

The IUPAC system aims to help chemists communicate more easily in all languages by introducing systematic names into compounds. However, its success in achieving this will be determined by how readily it is adopted. What do you think might prevent chemists using exclusively the 'new names'?



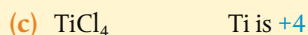
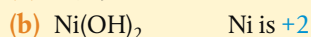
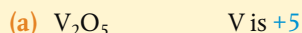
Worked example

Deduce the name of the following compounds using oxidation numbers.

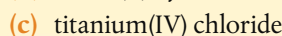
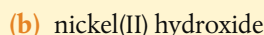
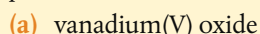


Solution

1 First deduce the oxidation state:



2 Then the corresponding Roman numeral is inserted after the name of the element. There is no space between the name and the number, and the number is placed in brackets.



Although this nomenclature can theoretically be used in the naming of all compounds, it is really only worthwhile when an element has more than one common oxidation state. For example Na_2O could be called sodium(I) oxide, but as we know Na always has oxidation state +1, it is perfectly adequate to call it simply sodium oxide.



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Oxidation state and formal charge (introduced in Chapter 4) are both useful tools for electron book-keeping, but they are based on very different premises. Oxidation state is derived by assuming bonds are ionic unless they are between the same element; formal charge is assigned by assuming bonds are pure covalent. Each model has its usefulness and its limitations, as discussed on pages 193 and 415.

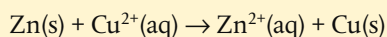
Redox equations

Writing half-equations

Although we have seen that oxidation cannot take place without reduction and vice versa, it is sometimes useful to separate out the two processes from a redox equation and write separate equations for the oxidation and reduction processes. These are therefore called **half-equations**. Electrons are added on one side of each equation to balance the charges.

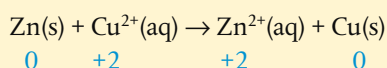
Worked example

Deduce the two half-equations for the following reaction



Solution

Assign oxidation states so we can see what is being oxidized and what is reduced.



Here we can see that Zn is being oxidized and Cu^{2+} is being reduced.

oxidation: $\text{Zn(s)} \rightarrow \text{Zn}^{2+}(\text{aq}) + 2\text{e}^-$ **electrons are lost**

reduction: $\text{Cu}^{2+}(\text{aq}) + 2\text{e}^- \rightarrow \text{Cu(s)}$ **electrons are gained**

Note there must be equal numbers of electrons in the two half-equations, so that when they are added together the electrons cancel out.

Writing redox equations using half-equations

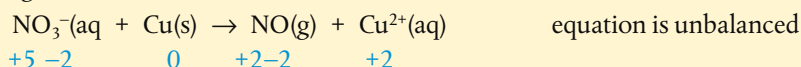
Sometimes we may know the species involved in a redox reaction, but not the overall equation so we need to work this out, making sure it is balanced for both atoms and charge. A good way to do this is to write half-equations for the oxidation and reduction processes separately, and then add these two together to give the overall reaction. Many of these reactions take place in acidified solutions and we therefore use H_2O and/or H^+ ions to balance the half-equations. The process is best broken down into a series of steps, as shown in the example below.

Worked example

Write an equation for the reaction in which NO_3^- and Cu react together in acidic solution to produce NO and Cu^{2+} .

Solution

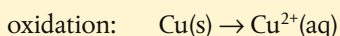
- 1 Assign oxidation states to determine which atoms are being oxidized and which are being reduced:



Therefore Cu is being oxidized ($0 \rightarrow +2$) and N is being reduced ($+5 \rightarrow +2$).

- 2 Write half-equations for oxidation and reduction.

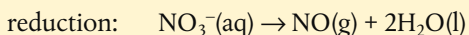
- (a) Balance the atoms other than H and O:



In this example the Cu and N are already balanced.

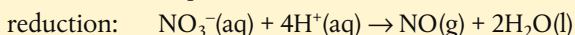
- (b) Balance each half-equation for O by adding H_2O as needed.

Here the reduction equation needs two more O atoms on the right-hand side:

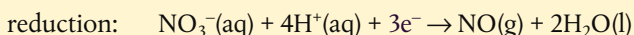
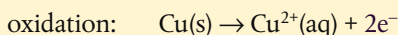


- (c) Balance each half-equation for H by adding H^+ as needed.

Here the reduction equation needs four H atoms on the left-hand side:



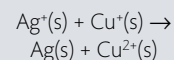
- (d) Balance each half-equation for charge by adding electrons to the side with the more positive charge. (Electrons will be products in the oxidation equation and reactants in the reduction equation.)



- (e) Now check that each half-equation is balanced for atoms and for charge.



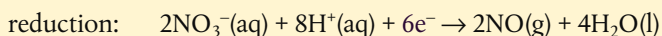
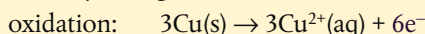
Photochromic lenses darken in response to increasing light intensity and become lighter in dim light. The change is brought about by a redox reaction between silver chloride and copper(I) chloride, which are both embedded in the lens. In bright light Ag^+ ions oxidize Cu^+ forming Ag and Cu^{2+} :



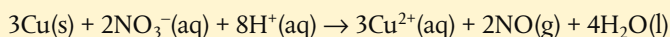
The silver metal produced creates a film over the lens, cutting out glare. In dim light the reaction reverses as Cu^{2+} oxidizes Ag, and the lens clears.

- 3** Equalize the number of electrons in the two half-equations by multiplying each appropriately.

Here the equation of oxidation must be multiplied by 3, and the equation of reduction by 2, to give six electrons in both equations:



- 4** Add the two half-equations together, cancelling out anything that is the same on both sides, which includes the electrons.



The final equation should be balanced for atoms and charge and have no electrons.



Summary of steps in writing redox equations.

- 1** Assign oxidation states to determine which atoms are being oxidized and which are being reduced.
- 2** Write half-equations for oxidation and reduction as follows:
 - (a) balance the atoms other than H and O;
 - (b) balance each half-equation for O by adding H_2O as needed;
 - (c) balance each half-equation for H by adding H^{+} as needed;
 - (d) balance each half-equation for charge by adding electrons to the sides with the more positive charge.
 - (e) check that each half-equation is balanced for atoms and for charge.
- 3** Equalize the number of electrons in the two half-equations by multiplying each appropriately.
- 4** Add the two half-equations together, cancelling out anything that is the same on both sides.

Exercises

- 1** Assign oxidation states to all elements in the following compounds:

(a) NH_4^{+}	(d) SO_2	(g) MnO_2	(i) $\text{K}_2\text{Cr}_2\text{O}_7$
(b) CuCl_2	(e) Fe_2O_3	(h) PO_4^{3-}	(j) MnO_4^{-}
(c) H_2O	(f) NO_3^{-}		
- 2** Use oxidation states to deduce which species is oxidized and which is reduced in the following reactions:
 - (a) $\text{Sn}^{2+}(\text{aq}) + 2\text{Fe}^{3+}(\text{aq}) \rightarrow \text{Sn}^{4+}(\text{aq}) + 2\text{Fe}^{2+}(\text{aq})$
 - (b) $\text{Cl}_2(\text{aq}) + 2\text{NaBr}(\text{aq}) \rightarrow \text{Br}_2(\text{aq}) + 2\text{NaCl}(\text{aq})$
 - (c) $2\text{FeCl}_2(\text{aq}) + \text{Cl}_2(\text{aq}) \rightarrow 2\text{FeCl}_3(\text{aq})$
 - (d) $2\text{H}_2\text{O(l)} + 2\text{F}_2(\text{g}) \rightarrow 4\text{HF(aq)} + \text{O}_2(\text{g})$
 - (e) $\text{I}_2(\text{aq}) + \text{SO}_3^{2-}(\text{aq}) + \text{H}_2\text{O(l)} \rightarrow 2\text{I}^{-}(\text{aq}) + \text{SO}_4^{2-}(\text{aq}) + 2\text{H}^{+}(\text{aq})$
- 3** Deduce the half-equations of oxidation and reduction for the following reactions:
 - (a) $\text{Ca(s)} + 2\text{H}^{+}(\text{aq}) \rightarrow \text{Ca}^{2+}(\text{aq}) + \text{H}_2(\text{g})$
 - (b) $2\text{Fe}^{2+}(\text{aq}) + \text{Cl}_2(\text{aq}) \rightarrow 2\text{Fe}^{3+}(\text{aq}) + 2\text{Cl}^{-}(\text{aq})$
 - (c) $\text{Sn}^{2+}(\text{aq}) + 2\text{Fe}^{3+}(\text{aq}) \rightarrow \text{Sn}^{4+}(\text{aq}) + 2\text{Fe}^{2+}(\text{aq})$
 - (d) $\text{Cl}_2(\text{aq}) + 2\text{Br}^{-}(\text{aq}) \rightarrow 2\text{Cl}^{-}(\text{aq}) + \text{Br}_2(\text{aq})$
- 4** Write balanced equations for the following reactions that occur in acidic solutions:
 - (a) $\text{Zn(s)} + \text{SO}_4^{2-}(\text{aq}) \rightarrow \text{Zn}^{2+}(\text{aq}) + \text{SO}_2(\text{g})$
 - (b) $\text{I}^{-}(\text{aq}) + \text{HSO}_4^{-}(\text{aq}) \rightarrow \text{I}_2(\text{aq}) + \text{SO}_2(\text{g})$
 - (c) $\text{NO}_3^{-}(\text{aq}) + \text{Zn(s)} \rightarrow \text{NH}_4^{+}(\text{aq}) + \text{Zn}^{2+}(\text{aq})$
 - (d) $\text{I}_2(\text{aq}) + \text{OCl}^{-}(\text{aq}) \rightarrow \text{IO}_3^{-}(\text{aq}) + \text{Cl}^{-}(\text{aq})$
 - (e) $\text{MnO}_4^{-}(\text{aq}) + \text{H}_2\text{SO}_3(\text{aq}) \rightarrow \text{SO}_4^{2-}(\text{aq}) + \text{Mn}^{2+}(\text{aq})$

5 Which equation represents a redox reaction?

- A** $\text{NaOH(aq)} + \text{HNO}_3\text{(aq)} \rightarrow \text{NaNO}_3\text{(aq)} + \text{H}_2\text{O(l)}$
B $\text{Zn(s)} + 2\text{HCl(aq)} \rightarrow \text{ZnCl}_2\text{(aq)} + \text{H}_2\text{(g)}$
C $\text{CuO(s)} + 2\text{HCl(aq)} \rightarrow \text{CuCl}_2\text{(aq)} + \text{H}_2\text{O(l)}$
D $\text{MgCO}_3\text{(s)} + 2\text{HNO}_3\text{(aq)} \rightarrow \text{Mg(NO}_3)_2\text{(aq)} + \text{H}_2\text{O(l)} + \text{CO}_2\text{(g)}$

6 The oxidation state of chromium is the same in all the following compounds except:

- A** CrCl_3 **B** Cr_2O_3 **C** $\text{Cr}_2(\text{CO}_3)_3$ **D** CrO_3

7 Deduce the names of the following compounds, using their oxidation numbers:

- (a)** Cr_2O_3 **(c)** HNO_3 **(d)** HNO_2 **(e)** PbO_2
(b) CuCl

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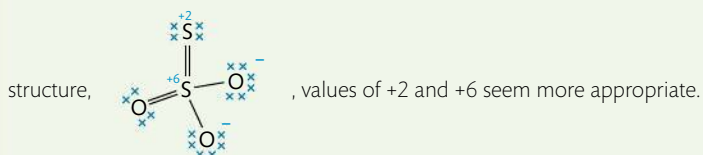
We have seen that oxidation states have several important uses as they enable us to:

- track redox changes in reactions;
- apply systematic names to compounds;
- determine reacting ratios in redox reactions.

At the same time, because they do not have a structural basis, oxidation states can raise some potential misunderstandings or ambiguities.

- Oxidation states do not distinguish between partial and complete transfer of electrons: +4 for C in CO_2 or +7 for Mn in KMnO_4 do not represent losses of 4 and 7 electrons respectively.
- In some cases, especially in organic chemistry, oxidation states have fractional values: in propane, C_3H_8 , the oxidation state of C is $-2\frac{2}{3}$.
- Oxidation states and Lewis (electron dot) structures sometimes suggest conflicting information: CH_4 , C_2H_6 , and C_3H_8 are all similarly bonded molecules, but their C atoms have different oxidation states, -4, -3, and $-2\frac{2}{3}$ respectively.

In $\text{S}_2\text{O}_3^{2-}$, the rules would suggest oxidation states of +2 on each S atom, but from the Lewis



Overall, oxidation states are best considered as a very useful tool that chemists use widely, while also keeping an eye on their limitations.

Oxidizing and reducing agents

We have seen that redox reactions *always* involve the simultaneous oxidation of one reactant with the reduction of another as electrons are transferred between them. The reactant that accepts electrons is called the **oxidizing agent** as it brings about oxidation of the other reactant. In the process it becomes reduced. Likewise the reactant that supplies the electrons is known as the **reducing agent**, because it brings about reduction and itself becomes oxidized. Sometimes the terms **reductant** and **oxidant** are used in place of reducing agent and oxidizing agent respectively.



Molten iron being tapped from a blast furnace for transfer to the steel-making furnace. Iron is extracted from its ore Fe_2O_3 by reducing it with carbon in the form of coke. Although blast furnaces existed in China from about the 5th century BCE and were widespread across Europe, a major development occurred in England in 1709. Substitution of the reducing agent charcoal for coke produced a less brittle form of iron. This accelerated the iron trade and was a key factor in the British Industrial Revolution.

In a redox equation the substance that is reduced is the oxidizing agent, the substance that is oxidized is the reducing agent.

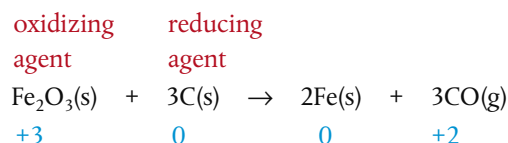
CHALLENGE YOURSELF

1 Hydrogen peroxide, H_2O_2 , can also act as an oxidizing agent and as a reducing agent. With reference to the oxidation states of its atoms, suggest why this is so and which behaviour is more likely.

Single replacement reactions involving metals with metal ions

Full details of how to carry out this experiment with a worksheet are available online.

For example, in the reaction where iron (Fe) is extracted from its ore (Fe_2O_3):



(The oxidation state of O is not shown as it does not change during the reaction.)

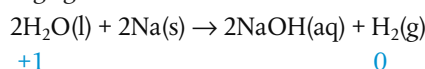
The reducing agent C brings about the reduction of Fe ($+2 \rightarrow 0$), while it is oxidized to CO ($0 \rightarrow +2$). The oxidizing agent Fe_2O_3 brings about the oxidation of C, while itself is reduced to Fe.

Some examples of useful oxidizing and reducing agents are given below:

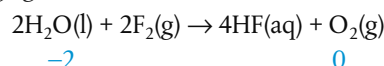
- oxidizing agent: O_2 , O_3 , H^+/MnO_4^- , $H^+/Cr_2O_7^{2-}$, F_2 , Cl_2 , conc. HNO_3 , H_2O_2
- reducing agent: H_2 , C , CO , SO_2 , reactive metals

Note that whether a species acts as an oxidizing or as a reducing agent actually depends on what it is reacting with. For example, water can act as an oxidizing agent and be reduced to hydrogen, for example by sodium, or act as a reducing agent and be oxidized to oxygen, for example by fluorine.

- H_2O acting as an oxidizing agent:



- H_2O acting as a reducing agent:



In fact, water is weak both as an oxidizing agent and as a reducing agent, which is why it is such a useful solvent for many redox reactions.

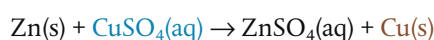


Treatment of drinking water to kill pathogens commonly uses oxidizing agents such as chlorine and ozone. The process is effective and relatively cheap, yet millions of people worldwide lack access to clean drinking water. 'Access to safe water is a fundamental human need and therefore a human right.' Kofi Annan, United Nations Secretary General. In 2010 this human right was passed into international law, but this does not yet make it a reality for millions of people in developing countries.

More reactive metals are stronger reducing agents

Of course not all oxidizing and reducing agents are of equal strength. Some will be stronger than others, depending on their relative tendencies to lose or gain electrons. We learned in Chapter 4 that metals have a tendency to lose electrons and form positive ions, so they will act as reducing agents, pushing their electrons on to another substance. More reactive metals lose their electrons more readily and so we might expect that they will be stronger reducing agents than less reactive metals.

We can check this out by seeing if one metal is able to reduce the ions of another metal in solution. If we immerse zinc in a solution of copper sulfate, a reaction occurs. The blue colour of the solution fades, the pinkish-brown colour of copper metal appears and there is a rise in temperature. What is happening is that the Cu^{2+} ions are being displaced from solution as they are reduced by Zn. At the same time Zn dissolves as it is oxidized to Zn^{2+} .



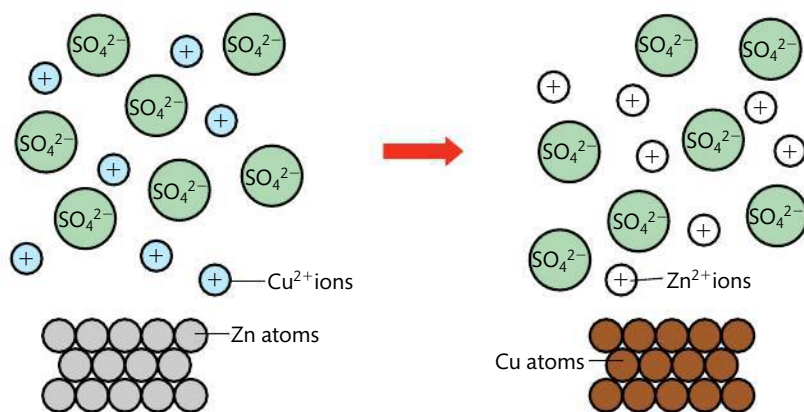
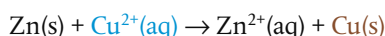


Figure 9.1 The reaction of zinc with copper(II) sulfate solution.

We can write this as an ionic equation without showing the sulfate ions as these act as spectator ions and are not changed during the reaction.



So zinc has acted as the reducing agent – it is therefore the more reactive metal. We can think of it as having the reducing strength to ‘force’ copper ions to accept the electrons.

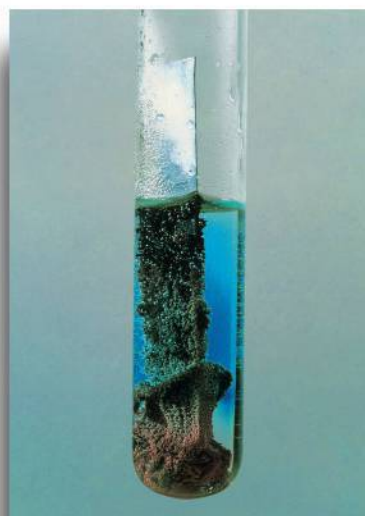
We could confirm this by trying the reaction the other way round, with copper metal immersed in a solution of zinc ions. Can you predict what will happen? The answer is there will be no reaction, because Cu is not a strong enough reducing agent to reduce Zn^{2+} . This is another way of saying that it is a less reactive metal, less able to push the electrons onto Zn^{2+} .

By comparing **displacement** reactions like these between different combinations of metals and their ions, we can build up a list of relative strengths of the metals as reducing agents. This is called the **activity series** and it enables us to predict whether a particular redox reaction between a metal and the ions of another metal will be feasible. Later in this chapter we will learn how to quantify these differences in metal reactivity, which is of great importance in many industrial processes. For example, the extraction of a metal from its ore often involves choosing a suitable reducing agent by reference to these data.

Here is a small part of the activity series of metals:

Mg	strongest reducing agent, most readily becomes oxidized
Al	
Zn	
Fe	
Pb	
Cu	
Ag	weakest reducing agent, least readily becomes oxidized

A simple activity series is given in section 25 of the IB data booklet and is not something that needs to be learned. The important thing is that you are able to interpret it and to deduce it from given data.



A strip of zinc metal half submerged in a solution of CuSO_4 . Solid copper that appears brown is deposited and the blue colour fades as the copper ions are reduced by the zinc. Zinc is shown to be the stronger reducing agent.



More reactive metals are stronger reducing agents than less reactive metals.



A more reactive metal is able to reduce the ions of a less reactive metal.

Redox reactions between a metal and the oxide of a less reactive metal can also confirm the activity series. Here powdered aluminium reduces iron oxide, forming aluminium oxide and iron.

$2\text{Al(s)} + \text{Fe}_2\text{O}_3\text{(s)} \rightarrow \text{Al}_2\text{O}_3\text{(s)} + 2\text{Fe(s)}$. This shows that aluminium is a stronger reducing agent than iron. The reaction is exothermic, creating short bursts of extremely high temperatures.



Worked example

Refer to the activity series given on page 417 to predict whether the following reactions will occur:

- (a) $\text{ZnCl}_2\text{(aq)} + 2\text{Ag(s)} \rightarrow 2\text{AgCl(s)} + \text{Zn(s)}$
- (b) $2\text{FeCl}_3\text{(aq)} + 3\text{Mg(s)} \rightarrow 3\text{MgCl}_2\text{(aq)} + 2\text{Fe(s)}$

Solution

- (a) This reaction would involve Ag reducing Zn^{2+} in ZnCl_2 . But Ag is a weaker reducing agent than Zn, so this will not occur.
- (b) This reaction involves Mg reducing Fe^{3+} in FeCl_3 . Mg is a stronger reducing agent than Fe, so this will occur.

We can investigate how some non-metals such as carbon and hydrogen would fit into this activity series of metals by similar types of displacement reactions. Carbon is able to reduce the oxides of iron and metals below it in the series, which provides one of the most effective means for the extraction of these metals. The position of hydrogen relative to the metals is discussed in section 19.1 on pages 432–435.

More reactive non-metals are stronger oxidizing agents

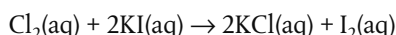
In a similar way, the different strengths of non-metals as oxidizing agents can be compared. For example, the halogens (Group 17 elements) react by gaining electrons and forming negative ions, and so act as oxidizing agents by removing electrons from other substances. We learned in Chapter 3 that their tendency to do this decreases down the group, so we would expect the following trend.

More reactive non-metals are stronger oxidizing agents than less reactive non-metals.

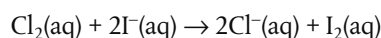


F	strongest oxidizing agent, most readily becomes reduced
Cl	
Br	
I	weakest oxidizing agent, least readily becomes reduced

Again this can be verified by reacting one halogen with solutions containing the ions of another halogen (known as halide ions). For example:



Here the Na^+ ions are spectator ions so we can write the ionic equation without showing them:



The reaction occurs because Cl is a stronger oxidizing agent than I and so is able to remove electrons from it. In simple terms, you can think of it as a competition for electrons where the stronger oxidizing agent, in this case chlorine, will always 'win'.

Exercises

- 8 Identify the oxidizing agents and the reducing agents in the following reactions:
- (a) $\text{H}_2(\text{g}) + \text{Cl}_2(\text{g}) \rightarrow 2\text{HCl}(\text{g})$
 - (b) $2\text{Al}(\text{s}) + 3\text{PbCl}_2(\text{s}) \rightarrow 2\text{AlCl}_3(\text{s}) + 3\text{Pb}(\text{s})$
 - (c) $\text{Cl}_2(\text{aq}) + 2\text{KI}(\text{aq}) \rightarrow 2\text{KCl}(\text{aq}) + \text{I}_2(\text{aq})$
 - (d) $\text{CH}_4(\text{g}) + 2\text{O}_2(\text{g}) \rightarrow \text{CO}_2(\text{g}) + 2\text{H}_2\text{O}(\text{l})$
- 9 Use the two reactivity series given to predict whether reactions will occur between the following reactants, and write equations where relevant.
- (a) $\text{CuCl}_2 + \text{Ag}$
 - (b) $\text{Fe}(\text{NO}_3)_2 + \text{Al}$
 - (c) $\text{NaI} + \text{Br}_2$
 - (d) $\text{KCl} + \text{I}_2$
- 10 (a) Use the following reactions to deduce the order of reactivity of the elements W, X, Y, Z, putting the most reactive first.
- $$\text{W} + \text{X}^+ \rightarrow \text{W}^+ + \text{X}$$
- $$\text{X} + \text{Z}^+ \rightarrow \text{X}^+ + \text{Z}$$
- $$\text{Y}^+ + \text{Z} \rightarrow \text{no reaction}$$
- $$\text{X} + \text{Y}^+ \rightarrow \text{X}^+ + \text{Y}$$
- (b) Which of the following reactions would you expect to occur according to the reactivity series you established in part (a)?
- (i) $\text{W}^+ + \text{Y} \rightarrow \text{W} + \text{Y}^+$
 - (ii) $\text{W}^+ + \text{Z} \rightarrow \text{W} + \text{Z}^+$



Chlorine gas bubbling through a clear solution of potassium iodide KI. The solution is turning brown due to the formation of iodine in solution, as chlorine oxidizes the iodide ions and forms chloride ions.



A more reactive non-metal is able to oxidize the ions of a less reactive non-metal.

Redox titrations

As with acid–base titrations (discussed in Chapter 8), redox titrations are used to determine the unknown concentration of a substance in solution. But in this case, the technique is based on a redox reaction between the two reactants, and finds the equivalence point where they have reacted stoichiometrically by transferring electrons. The two types of titration are compared below.

Acid–base titration	Redox titration
neutralization reaction between acid and base	redox reaction between oxidizing agent and reducing agent
protons are transferred from acid to base	electrons are transferred from reducing agent to oxidizing agent

Redox titrations are carried out in much the same way as acid–base titrations, using a burette and pipette to measure volumes accurately, and a standard solution of

CHALLENGE YOURSELF

- 2 Reactions in which the same element is simultaneously oxidized and reduced is known as **disproportionation**. Show how this happens in the reaction when Cl_2 and NaOH react to produce NaCl , NaClO , and H_2O .



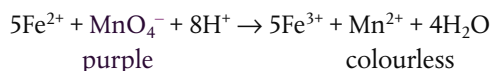
The browning of fruit when it is exposed to the air is an oxidation reaction. It occurs when phenols released in damaged cells are oxidized with molecular oxygen at alkaline pH. The brown products are tannins. Food chemists are often interested in slowing down these reactions using reducing agents such as SO_2 or vitamin C, for example in lemon juice.

one reactant. An indicator is usually used to signal the equivalence point, although some redox changes are accompanied by a colour change and may not need an external indicator. From the volume of the solution added from the burette to reach equivalence, which is known as the **titre**, the concentration of the other reactant can be determined. The calculations are based on the redox equation, which is typically developed from the half-equations.

Redox titrations are commonly used in the food and beverages industry, in the pharmaceutical industry, and in water and environmental analysis. For example, wines can be analysed for the presence of sulfur dioxide, and the vitamin C content of foods can be determined in this way. Some examples of common redox titrations that can be carried out in the laboratory are given below, and more details on how to carry out these experiments can be found online.

1 Analysis of iron with manganate(VII)

This redox titration uses KMnO_4 in an acidic solution as the oxidizing agent, which oxidizes Fe^{2+} ions to Fe^{3+} . During the reaction MnO_4^- is reduced to Mn^{2+} , so the overall equation is:



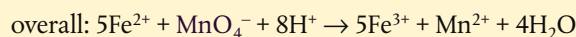
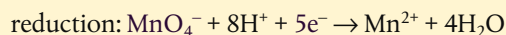
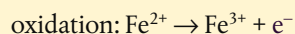
The reaction is accompanied by a colour change from deep purple to colourless, so the reaction acts as its own indicator, signalling the equivalence point.

Worked example

All the iron in a 2.000 g tablet was dissolved in an acidic solution and converted to Fe^{2+} . This was then titrated with KMnO_4 . The titration required 27.50 cm^3 of $0.100 \text{ mol dm}^{-3} \text{ KMnO}_4$. Calculate the total mass of iron in the tablet and its percentage by mass. Describe what would be observed during the reaction, and how the equivalence point can be detected.

Solution

First we need the balanced equation for the reaction, which is solved by the half-equation method described earlier.

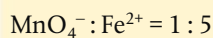


Next we need to know the amounts of reactants used to reach equivalence. We start with KMnO_4 as we know both its concentration and its volume.

$$n = cV$$

$$n(\text{MnO}_4^-) = 0.100 \text{ mol dm}^{-3} \times \frac{27.50}{1000} \text{ dm}^3 = 0.00275 \text{ mol MnO}_4^-$$

From the equation for the reaction we know the reacting ratio:



$$\therefore \text{mol Fe}^{2+} = 0.00275 \text{ mol MnO}_4^- \times 5 = 0.01375 \text{ mol}$$

Redox titration

Full details of how to carry out this experiment with a worksheet are available online.



$$M(\text{Fe}) = 55.85 \text{ g mol}^{-1}$$

$$n = \frac{m}{M}$$

$$\therefore m(\text{Fe}) = 0.01375 \text{ mol} \times 55.85 \text{ g mol}^{-1} = 0.7679 \text{ g}$$

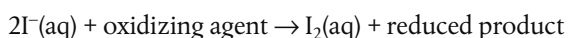
$$\% \text{ Fe in tablet} = \frac{0.768 \text{ g}}{2.000 \text{ g}} \times 100 = 38.39\%$$

$$\therefore \text{Fe in tablet} = 0.768 \text{ g}, 38.4\%$$

MnO_4^- in the burette is purple, but forms a nearly colourless solution in the flask as it reacts to form Mn^{2+} . But when the reducing agent Fe^{2+} in the flask has been used up at equivalence, MnO_4^- ions will not react and the purple colour will persist.

2 Iodine–thiosulfate reaction

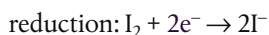
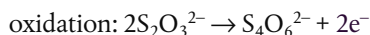
Several different redox titrations use an oxidizing agent to react with excess iodide ions to form iodine.



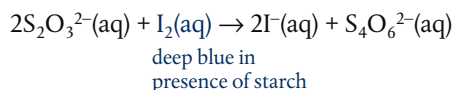
Examples of oxidizing agents used in this way include KMnO_4 , KIO_3 , $\text{K}_2\text{Cr}_2\text{O}_7$, and NaOCl .

The liberated iodine, I_2 , is then titrated with sodium thiosulfate, $\text{Na}_2\text{S}_2\text{O}_3$, using starch as an indicator.

Redox equations:



The overall equation is:



The starch indicator is added during the titration (not at the start) and forms a deep blue colour by forming a complex with free I_2 . As the I_2 is reduced to I^- during the reaction, the blue colour disappears, marking the equivalence point.

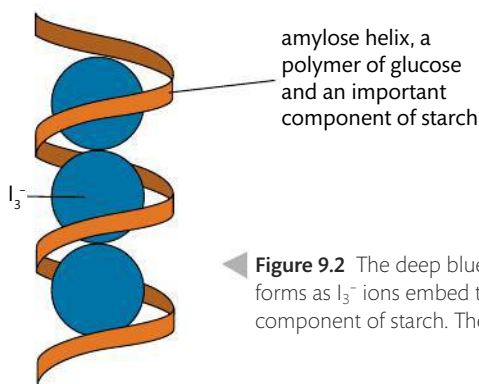


Figure 9.2 The deep blue complex between iodine solution and starch forms as I_3^- ions embed themselves within the helix of amylose, a component of starch. The structure of I_3^- was discussed in Chapter 4.



Because iodine, I_2 , has a very low solubility in water, it is used instead in the form of 'iodine solution', which is made by dissolving I_2 in iodide ions, forming the complex ion $\text{I}_3^-(\text{aq})$. It is the I_3^- ion that gives the deep blue complex with starch. Even though I_3^- is the reactant, it is still valid to use I_2 in the equations, as this is the 'active ingredient'. Technically, starch is not a redox indicator as such, as it responds to the disappearance of I_2 , rather than to a change in the redox potential of the solution.

CHALLENGE YOURSELF

- 3** From your knowledge of the structure of I_3^- , consider why it forms a stable complex in the hydrophobic interior of the amylose helix.

In titration calculations, always look first for the reactant where you know both its reacting volume and its concentration. Start by using these data to calculate the number of moles. Then use the reacting ratio in the equation to deduce the moles of the other reactant.

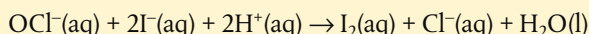


A canal in England covered in a bloom of duckweed. This is an aquatic weed that thrives in water that contains pollutants such as phosphates. This is known as eutrophication. The bloom blocks out sunlight to deeper layers of the water ecosystem, and so lowers the level of dissolved oxygen.



Worked example

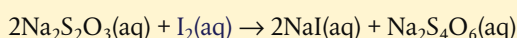
Household bleach is an oxidizing agent that contains sodium hypochlorite, NaOCl, as the active ingredient. It reacts with iodide ions in acidic solution as follows:



A 10.00 cm³ sample of bleach was reacted with a solution of excess iodide ions, and the liberated iodine was then titrated with Na₂S₂O₃. The titration required 38.65 cm³ of 0.0200 mol dm⁻³ Na₂S₂O₃. Determine the concentration of OCl⁻ in the bleach.

Solution

First we need the balanced equation for the titration, which is solved by the half-equation method on page 413.

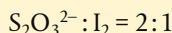


Next we calculate the number of moles of Na₂S₂O₃ as we know both its concentration and its volume.

$$n = cV$$

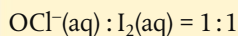
$$n(\text{S}_2\text{O}_3^{2-}) = 0.0200 \text{ mol dm}^{-3} \times \frac{38.65}{1000} \text{ dm}^3 = 7.73 \times 10^{-4} \text{ mol S}_2\text{O}_3^{2-}$$

From the equation for the reaction we know the reacting ratio



$$\therefore \text{mol I}_2 = 7.73 \times 10^{-4} \times 0.5 = 3.865 \times 10^{-4}$$

This is a back-titration as the I₂ was liberated by the reaction of I⁻ with OCl⁻, as given in the question. The reacting ratio is



$$\therefore \text{mol OCl}^- = \text{mol I}_2 = 3.865 \times 10^{-4}$$

The concentration of the OCl⁻ in the bleach can now be calculated.

$$n(\text{OCl}^-) = cV$$

$$\therefore 3.865 \times 10^{-4} \text{ mol} = c \times \frac{10.00}{1000} \text{ dm}^3$$

$$[\text{OCl}^-] = 0.0387 \text{ mol dm}^{-3}$$

3 Winkler method for calculating dissolved oxygen

The **dissolved oxygen** content of water is one of the most important indicators of its quality, as oxygen is essential to the survival of aquatic life. As the level of pollution in water increases, the dissolved oxygen content generally decreases, as the oxygen is used by bacteria in decomposition reactions. The **biological oxygen demand** (BOD) is therefore used as a means of measuring the degree of pollution. BOD is defined as the amount of oxygen used to decompose the organic matter in a sample of water over a specified